

1-10-2026

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT

Decode the Chart

Name: _____ K. Deeksha Sree _____ Roll No: _____ 16 _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

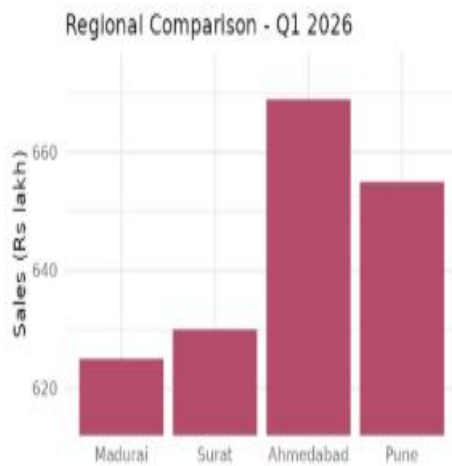


Chart 1

Market Share by Product (11 categories)



Chart 2

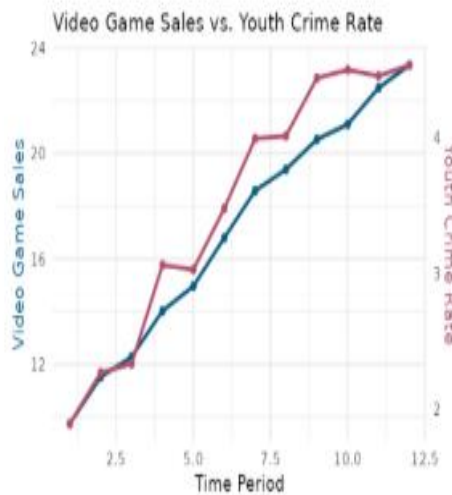


Chart 3

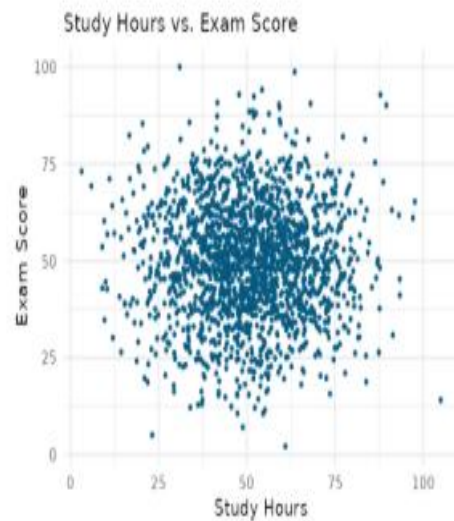


Chart 4

1-10-2026

Your notes:

Chart 1 — Truncated y-axis (bar chart)

Chart 1 shows a bar chart comparing sales across **Madurai, Surat, Ahmedabad, and Pune**. The y-axis starts around **620 instead of 0**, which makes the relatively small differences in sales appear much larger than they actually are. For example, Ahmedabad appears dramatically higher than Madurai, even though the actual difference is only a few tens of lakhs.

Rule: Bar charts should normally start at zero because the length of each bar represents the magnitude of the value. Starting the y-axis above zero can visually exaggerate differences. This violates the **Principle of Proportional Ink**.

Chart 2 — Too many categories in a pie chart

Chart 2 is a pie chart showing **market share across 11 product categories**. With 11 wedges, several slices are difficult to distinguish and compare accurately because their angles and areas are very similar. The many colors also make it harder to identify which product has the largest or smallest share.

Rule: Pie charts work best with a small number of categories, usually around **6–8 or fewer**. When there are many categories, use a **sorted bar chart with direct labels** so the values can be compared more accurately.

Chart 3 — Dual y-axis and misleading visual correlation

Chart 3 compares **Video Game Sales** and **Youth Crime Rate** using two different y-axes. The two lines appear to move together at several points, but they are measured on different scales and represent completely different quantities. Changing the scale of either axis can make the lines appear more or less correlated without changing the underlying data.

Rule: Be cautious with **dual-axis charts**, especially when suggesting relationships between unrelated variables. Prefer a **single shared scale, separate panels, or a scatter plot with a correlation statistic**. A visual overlap alone does not establish a real relationship.

Chart 4 — Overplotting (scatter plot)

Chart 4 shows **Study Hours vs. Exam Score** with a very large number of points. Because many points are fully opaque and overlap in the central region, the dense areas appear as solid clusters, making it difficult to see the actual distribution and concentration of observations. Individual observations are also difficult to distinguish.

Rule: For large scatter-plot datasets, use **partial transparency (alpha)**, smaller points, **2D histograms/hex bins**, or **contour/density plots**. These techniques make the underlying data distribution easier to interpret.